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PsychSupport – Student Mental Health Support Platform

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1. Introduction

1.1.1. Motivation

The contemporary academic environment imposes immense psychological pressure, examination stress, and adaptive challenges on university students, severely affecting their mental health. Traditional campus psychological support systems often face significant barriers, including long waiting times and the social stigma associated with seeking help. The primary motivation behind developing **PsychSupport** was to design a low-barrier digital tool capable of delivering immediate mental health support within a single, cohesive web interface.

The core of this project is its *anonymity-first* approach. A student in acute distress can immediately access evidence-based calming techniques, sensory grounding exercises, and crisis contacts without creating an account or revealing their identity. For deeper engagement, a graduated access model allows students to register with their university email to unlock advanced features like peer-support chatbot interactions, an educational library, and appointment booking with campus therapists.

1.1.2. Project Team

The PsychSupport platform was designed and developed by a dedicated three-person student project team. Operating as a cohesive unit within an academic framework, the team blended competencies in full-stack development, user interface design, and project management to take the platform from initial concept to a deployment-ready prototype.

Responsibilities were shared across the application's lifecycle, requiring close collaboration to bridge frontend layout prototyping, database schema integration, and technical documentation. By working collectively, the team ensured that all architectural, security, and design choices remained strictly aligned with the overarching mission of creating a safe, low-barrier, and student-centered mental health resource.

1.1.3. Assumed Project Methodology

To manage the development lifecycle within an academic timeframe, the team used an **Agile/Scrum** inspired methodology. Work was structured into short, iterative sprints for rapid feature implementation. The initial frontend skeleton and UI components were generated using the **v0** AI tool, which significantly accelerated the prototyping phase. From there, the team built out separate application modules (Home, Chat, Calendar, Materials), implemented secure Next.js Server Actions, and embedded client-side keyword-based rules for real-time crisis detection in the chat interface.

1.1.4. Description of Company / Environment

The target institutional environment for this application is the **University Administration and Student Welfare Department**. Higher education institutions operate under strict student privacy constraints, GDPR compliance, and limited IT maintenance budgets.

2. Section Project description

2.1.1. Functional Modules and User Access Levels

The platform is built around a graduated access model designed to lower the initial barrier for students seeking mental health care. The application establishes two distinct user tiers to balance immediate privacy with data-driven features:

- **Anonymous Access Layer:** Requires no registration or personal data. Upon launching the root page (`app/page.tsx`), a database trigger automatically pairs the session with a randomized, human-readable identifier (e.g., ANON-XK7F2A). This state grants immediate access to the *Home (Calming Tips)* panel and the *SOS Emergency Dialog*.
- **Registered User Layer:** To access advanced interactive features, students register using a verified university email, phone number, and a 6-digit student ID. This unlocks personalized components including the therapist booking calendar, peer-support chat, and the psychoeducational library.

2.1.2. Detailed Breakdown of Core Platform Features

The application coordinates four main domain views alongside a global emergency overlay managed via client-side state:

- **Home – Calming Tips:** Displays 8 cards featuring stress-coping techniques (e.g., Box Breathing, Grounding). Clicking a card opens a modal window containing step-by-step instructions and an explanation of the underlying biological mechanism (e.g., parasympathetic nervous system activation).
- **Anonymous Peer-Support Chatbot:** A fully client-side assistant. It evaluates the emotional tone of user messages (including crisis states) using a keyword detection engine and selects appropriate responses from localized dictionaries.
- **Therapist Appointment Booking Calendar:** A weekly schedule (Mon.–Sat., 8:00 AM–8:00 PM) powered by real-time data from Supabase. Time slots are color-coded: blue for online consultations and green for on-campus sessions. The booking workflow allows the user to select a therapist and automatically generates a video meeting link or provides the physical office address.
- **Pre-Appointment Support Cards:** An initial assessment form for on-campus visits. It measures mood and stress levels on a scale from 1 to 10 and gathers a brief description of the user's challenges and goals. The card is shared with the therapist prior to the session to maximize its effectiveness.
- **Psychoeducational Knowledge Library:** A search tool for student wellness articles. The content is embedded directly within the component, eliminating database latency during browsing.
- **SOS Emergency Dialog:** A 24/7 accessible red modal window located in the sidebar, providing direct-dial links (tel:) to crisis hotlines (116 123, 112) and SMS support. Upon detecting crisis-related keywords, the chatbot automatically updates the interface and redirects the user to this window.

2.1.3. Full-Stack Architecture and Rendering Strategy

The platform functions as a single-page architecture nested within the Next.js App Router framework, prioritizing responsive client-side view rendering while delegating data mutations

strictly to the server. As shown in the codebase file tree, all project configs reside at the root level. The architecture operates via three execution layers:

- **Edge Middleware (`middleware.ts`):** Intercepts every incoming network request to evaluate and update the Supabase session token stored inside encrypted HTTP-only cookies, checking validity directly against the server client.
- **Client Shell (`app/page.tsx`):** Coordinates initial application bootstrapping, listening for authentication state changes and swapping between the AuthView and the main Dashboard layout.
- **Server Actions API (`app/actions/`):** Isolates backend logic from the browser. The `auth.ts` file manages synthetic account generation for anonymous profiles and returning user validation. The `appointments.ts` file handles live availability slot-aggregation loops and direct table data injections.

2.1.4. Inferred Database Schema Structure

The backend utilizes a relational Supabase BaaS layer consisting of five key tables inferred directly from server-side data access queries:

- **profiles:** Tracks primary user UUIDs, trigger-generated anonymous session IDs, and student profile metadata.
- **therapists:** A directory of active university counselors storing titles, specialties, biographies, and overall peer ratings.
- **therapist_availability:** Logs weekly recurring operational blocks, storing days of the week, 24-hour start/end timestamps, and session delivery channels (online vs. offline).
- **appointments:** Stores finalized booking records linking student IDs to specific therapist slots, calendar dates, session notes, and auto-generated meeting hyperlinks.
- **support_cards:** Retains the quantitative mood ratings and textual challenges mapped directly to their corresponding appointment entries.

2.1.5. Technical Specifications and Governance

- **UI System & Styling:** Employs Tailwind CSS v4 alongside `shadcn/ui` (New-York variant), relying on Radix UI primitives for built-in accessibility compliance. Design tokens are maintained as custom CSS properties inside `app/globals.css` rather than an external configuration file.
- **TypeScript Configuration:** Configured under TypeScript 5.7 strict mode, utilizing incremental caching strategies and root-level path aliases (`@/*`) for clean import compilation.
- **Compliance & Analytics:** Includes a `CookieConsentBanner` component that saves user tracking preferences (necessary, analytics, marketing) to local storage. External production packages, such as Vercel Analytics, are explicitly isolated to execute only within true production environments.

2.2. Team members

- **Joanna Joniec (Project Lead):** Handled the overall project vision, shaping PsychSupport into a student-centered solution and aligning functionalities with user needs.

- **Marcin Gronek (Technical Development):** Focused on full-stack implementation, backend architecture, Supabase database integration, and session security.
- **Szymon Gustek (Product Design & Communication):** Managed the user interface layout, translating mental health workflows into an accessible and clean user experience.

3. Section Business plan

Financial Framework and Resource Optimization

The financial strategy of the PsychSupport platform is explicitly structured around the modern **Serverless/BaaS (Backend-as-a-Service)** model. Unlike traditional enterprise platforms that require substantial upfront investments in physical infrastructure, this architecture allows the target university to operate with near-zero initial infrastructure overhead. The financial predictability and resource requirements are defined across the following three core items:

- **Zero Fixed Infrastructure Maintenance Cost** By leveraging a managed cloud backend via Supabase and static/edge hosting through Vercel, the application eliminates the need for dedicated physical hardware, server rooms, or continuous manual server maintenance. All database operations, authentication flows, and storage triggers run in a fully managed environment. This means the university's IT department avoids ongoing infrastructure tracking, manual hardware update cycles, or the financial burden of keeping an administrative hosting team on standby.
- **On-Demand Auto-Scalability During Peak Demand** The platform's financial model scales dynamically according to real-time student activity. During low-traffic periods, operational costs drop to baseline tiers. However, during predictable high-demand academic windows—such as the lead-up to examination sessions, final thesis deadlines, or winter/summer sessions—the serverless architecture scales up automatically to process increased traffic spikes. This system absorbs high workloads without requiring manual infrastructure adjustments or permanent, expensive over-provisioning of server resources.
- **Low Total Cost of Ownership (TCO) and High Budget Predictability** Because the platform minimizes external service dependencies and relies on native Next.js Server Actions rather than independent intermediate API layers, the Total Cost of Ownership is heavily optimized. Budget predictability remains highly precise, allowing university administrations with finite IT resources to forecast operational costs based directly on the active student population size. Financial expenditure is strictly tied to actual resource consumption (such as database reads/writes and secure session tokens), removing unexpected maintenance overheads.

3.1. Analysis of competitive solutions/analysis of market (mandatory)

The market for student mental health solutions is divided into traditional institutional care and commercial wellness applications, both of which present significant operational and psychological barriers that **PsychSupport** is specifically engineered to overcome.

- **Traditional University Counseling Services:** These legacy systems represent the primary baseline for campus care but require immediate, identifiable intake requests. This creates an immense psychological hurdle, causing students to avoid seeking help due to fear of stigma, academic judgment, or social stigmata. Additionally, they are often plagued by long waiting lists and fixed operational hours. *The PsychSupport Edge:* It bypasses this initial friction through its **Anonymity-First** design layer, automatically assigning a

randomized session identifier (e.g., ANON-XK7F2A) so students can access evidence-based calming techniques without disclosing any real-world identity.

- **Commercial Wellness and Meditation Apps (e.g., Headspace, Calm):** These profit-driven, consumer-facing applications lock core features behind expensive subscription paywalls and aggressively collect personal telemetry data. Furthermore, they operate in complete isolation from the student's actual university framework. *The PsychSupport Edge:* PsychSupport is a fully university-subsidized portal. It incorporates a strict, GDPR-compliant cookie consent manager that bars marketing trackers and isolates web analytics strictly to production environments. Most importantly, it actively bridges the gap to institutional resources through a permanent, visible SOS emergency dialog and a live calendar scheduling grid that directly connects students with active on-campus professionals.

By executing a **Graduated Access Model**, PsychSupport effectively combines the immediate, unauthenticated privacy of consumer utility tools with the trusted, safeguarded escalation pathways required by institutional university welfare standards.

3.2. Risk analysis (mandatory)

Implementing a digital mental health platform within a university environment requires managing technical, security, and operational challenges. Because the system is designed to connect students directly with human campus professionals and counselors, the risk management framework focuses on system availability, data privacy, and immediate escalation pathways.

Core Project Risks and Mitigation Actions

- **R-01: System Crashes During Peak Academic Stress Windows**
 - *Risk and Impact:* During high-stress institutional periods—such as final examinations or thesis submission deadlines—the volume of students seeking coping resources or trying to book scheduling slots rises sharply. Traditional physical infrastructure risks crashing under sudden traffic surges, leaving vulnerable students cut off from immediate care.
 - *Mitigation:* As shown in the directory structure of, the platform relies on a serverless framework combining Next.js 16 and a managed Supabase cloud database backend. This modern architecture scales system capacity up or down automatically on-demand, requiring zero physical server maintenance from the university IT team while guaranteeing continuous, high-availability access.
- **R-02: Exposure of Sensitive Student Identities or Interaction Records**
 - *Risk and Impact:* Student counseling files are highly sensitive; an accidental data leak or session-hijacking incident could compromise user anonymity, violate strict GDPR requirements, and damage institutional trust.

- *Mitigation:* For unauthenticated guest tiers, database triggers automatically generate randomized, meaningless identifiers (e.g., ANON-XK7F2A) completely detached from real-world personas. For registered users, all critical database operations are handled entirely server-side using secure Next.js Server Actions (app/actions/). Session tokens are isolated inside encrypted, HTTP-only browser cookies refreshed on every request by edge middleware (middleware.ts), completely blocking client-side script interference.
- **R-03: Operational Disconnect in Connecting Students to Campus Professionals**
 - *Risk and Impact:* Because the system integrates an active calendar to connect students with live human counselors, an operational risk occurs if a student requests urgent assistance outside of business hours or during high counselor caseloads.
 - *Mitigation:* To ensure students in critical distress never face system delays, the platform embeds a permanent, full-screen SOS Emergency dialog within the sidebar layout. This component runs independently of login state, providing instant access to direct-dial emergency telephone hotlines (116 123, 112). Operationally, the university must establish clear out-of-hours workflows and booking moderation protocols for the welfare team prior to full deployment.
- **R-04: Deployment of Unstable Code Due to Bypassed Strict Type Checks**
 - *Mitigation:* While this configuration accelerates early student prototyping and feature assembly, production hardening protocols dictate that this flag must be set to false prior to institutional deployment. This forces the build pipeline to strictly respect the TypeScript 5.7 strict mode constraints defined in tsconfig.json, ensuring absolute type safety across all dynamic appointments and authentication logic.

4. Details of proposed solution

4.1. System Shell and Single-Page User Experience

The core architecture of the PsychSupport platform is designed as a highly responsive single-page experience optimized for a fluid student user interface. By utilizing a singular mounting page shell, the application handles view transitions dynamically, completely eliminating standard browser loading latencies and page refreshes.

When a user visits the application, an initial onboarding gatekeeper layer automatically evaluates whether an active session exists. Based on this validation, the system conditionally displays either the onboarding screen or the main student dashboard. All primary navigation targets—including

the Home panel, Chat interface, Booking Calendar, and Materials section—are swapped seamlessly inside the central workspace via instant local interface updates. This client-centric navigation provides a frictionless experience that minimizes cognitive fatigue, ensuring that students seeking urgent psychological care do not face frustrating technical delays.

4.2. Dual-Tiered Graduated User Access Model

To effectively balance immediate student privacy with the administrative and tracking requirements of institutional health care, the platform utilizes a graduated access model. This approach splits user access into two distinct functional layers:

- **Anonymous Access Layer:** This tier acts as a zero-barrier digital front door for students experiencing high-stress states who are hesitant to reveal their identity. With a single click, the platform sets up a hidden, randomized secure account in the background without requiring the student to input any real credentials or personal information. The system automatically assigns a clean, human-readable identifier (such as ANON-XK7F2A). This temporary profile unlocks immediate, read-only entry to evidence-based calming guides and full interaction with the global SOS Emergency resources.
- **Registered User Layer:** When a student feels ready to engage with active university counseling staff, they can easily upgrade their account status. Full registration requires five standard inputs: first and last name, a university email address, an international-standard phone number, and a 6-digit student ID number. Once the student verifies their university email via a confirmation link, their profile status is upgraded. This unlocks interactive and data-driven features, including the live therapist booking calendar, personalized message history, pre-appointment mood diagnostics, and personal educational tracking tools.

4.3. Secure Data Flow and Session Protection

To maintain institutional trust, all data security and credential validations are managed securely behind the scenes. The platform coordinates independent security layers that govern the student's browser state, background server actions, and network routing.

Every time a user navigates within the platform, a network-layer middleware script refreshes their session token directly against the core authentication endpoints. By using encrypted, browser-hidden parameters (HTTP-only cookies), PsychSupport guards the active student session against malicious scripts, cross-site hijacking attempts, and sudden session expirations. If a session is tampered with or deemed invalid, access to sensitive student records or appointment scheduling is instantly blocked, ensuring absolute client confidentiality.

4.4. The Home Coping Module: Evidence-Based Calming Techniques

The Home panel serves as the central landing hub for all users. It features a responsive grid of eight interactive digital coping cards, each representing a scientifically backed psychological exercise tailored to reduce acute stress, panic, and dissociation. Clicking any card opens a detailed guidance window containing a step-by-step walkthrough and a plain-language explanation of its biological or clinical benefit:

- **Box Breathing:** Guides the student through a rhythmic 4-4-4-4 second breathing cycle, helping balance respiration rates to trigger the body's parasympathetic nervous system and lower elevated heart rates.
- **5-4-3-2-1 Grounding:** Instructs the user to identify five sights, four touches, three sounds, two smells, and one taste, effectively breaking acute panic loops through systematic sensory anchoring.
- **Progressive Muscle Relaxation:** Leads a head-to-toe somatic relaxation routine by sequentially contracting and releasing muscle groups.
- **Mindful Listening:** Prompts users to close their eyes and non-judgmentally isolate immediate environmental sounds, pulling attention away from internal anxious thoughts.
- **Self-Compassion Practice:** Focuses on constructive self-kindness exercises to mitigate harsh self-criticism following academic setbacks.
- **Safe Place Visualisation:** Uses guided mental imagery to establish an internal refuge, reducing active stress hormone levels.
- **Sleep Hygiene Optimization:** Provides a behavioral and environmental checklist to restore restorative sleep cycles during stressful exam periods.
- **Wave Breathing:** Uses an extended exhalation pattern (a 5:7 rhythm) to naturally stimulate the vagus nerve and ease physiological distress.

4.5. Interactive Peer-Support Chatbot Interface

The text communication workspace (ChatView) offers a warm, intuitive environment built to simulate real-time peer support. To ensure total conversation privacy, all message processing and emotional tone evaluations run completely inside the student's local browser window, meaning raw conversational text is never transmitted to outside servers.

As a student types, a client-side sentiment detection engine analyzes word choices and categorizes the emotional tone of the input text into one of five states:

1. **Crisis:** Triggers instantly if high-risk terms regarding self-harm or suicidal ideation are typed, immediately prioritizing a specialized response.
2. **Positive:** Identifies joyful or celebratory phrasing.
3. **Negative:** Flags distressed or self-critical sentence structures.
4. **Inquiry:** Recognizes active question-asking and help-seeking phrases.
5. **Neutral:** Acts as the baseline tone when no distinct emotional triggers dominate.

To preserve a comforting and natural flow, the chat interface simulates a human typing rhythm by varying the response delay between 1.0 and 2.5 seconds. It also tracks and rotates text options to prevent repetitive, robotic responses. While the student's unique anonymous identifier is displayed in the toolbar header for privacy reassurance, it is never transmitted alongside chat content.

4.6. Live Therapist Booking System

The scheduling module (CalendarView) seamlessly connects registered students with available university counseling staff using an interactive weekly grid. The grid maps available appointments from Monday through Saturday (8:00 AM to 8:00 PM). To ensure quick user recognition, time

slots are color-coded by consultation type: blue cells represent virtual online video consultations, and green cells denote stationary, face-to-face sessions on campus.

The booking workflow is managed through a reassuring three-phase path:

1. **Select Therapist Phase:** Clicking a slot queries active counselor timetables, presenting a list of available practitioners matching that hour. Each profile displays the counselor's credentials, bio, specializations, aggregate student rating index, and total reviews.
2. **Form Completion Phase:** The user's form adapts to the chosen session format. Virtual online bookings request the student's preferred language, primary topic, and notes. In-person sessions redirect the student to complete a Pre-Appointment Support Card.
3. **Confirmation Phase:** Upon successful scheduling, the interface locks in the appointment and provides an immediate summary. For online sessions, it provides a secure, randomized video chat room link. For on-campus sessions, it delivers the exact office room and physical building address of the practitioner.

4.7. Pre-Appointment Support Cards

The Support Cards module serves as an intentional clinical bridge between initial digital requests and face-to-face professional care. When a student schedules an on-campus visit, they fill out a brief digital context form that captures both quantitative and qualitative insights regarding their emotional health:

- **Quantitative Sliders:** Interactive touch sliders that allow students to rate their immediate mood levels and acute stress levels on a straightforward scale from 1 to 10.
- **Qualitative Blocks:** Open-ended text fields where students can naturally describe the recent challenges they are facing, along with their primary goals for the upcoming session.

Once finalized, the support card links directly to the corresponding appointment record. This vital context is made available to the assigned counselor prior to the session, enabling the professional to prepare thoroughly, spot urgent issues, and maximize session efficiency from the very first minute.

4.8. Psychoeducational Knowledge Library

The educational module (MaterialsView) serves as an independent, low-barrier learning space for students who may not be ready to schedule a formal counseling appointment but wish to explore mental health strategies on their own. The library provides a searchable, organized repository of wellness articles explicitly written for academic stress management, complete with topic badges and read times:

- Understanding Anxiety: A Student's Guide (Anxiety category; 8-min read).
- Building Resilience Through Self-Care (Self-Care category; 6-min read).
- Sleep Hygiene for Better Mental Health (Sleep category; 5-min read).
- Navigating Social Relationships (Relationships category; 7-min read).
- Managing Academic Stress (Stress category; 6-min read).
- Mindfulness for Beginners (Mindfulness category; 5-min read).

To completely eliminate network waiting delays, all article contents and metadata are embedded natively within the application layer, ensuring resources display instantly without needing to fetch data from an external server.

4.9. Global Safeguarding Net: The SOS Emergency Dialog

The `SosDialog` component provides an ironclad global safety net within the application. Positioned permanently within the layout's sidebar footer, it remains completely visible and clickable across all views, regardless of whether a user is logged in, browsing anonymously, or an unauthenticated guest.

Activating the button launches a full-screen, highly noticeable crisis dashboard with direct-dial links (tel: strings) optimized for mobile devices. It displays four paramount crisis lines:

1. **National Crisis Hotline:** Direct speed-dial to the 116 123 support helpline with a prominent 'Call Now' button.
2. **Emergency Services:** Instant contact numbers for the standard 112 medical/emergency dispatch.
3. **Crisis Text Support:** Clear instructions to text the keyword "HOME" to 741741 to instantly connect with a live crisis counselor over SMS.
4. **University Counselling Directory:** Direct instructions guiding the student on how to reach the university's on-call mobile mental health crisis team.

The system maintains an automated safety link with the interactive workspace. If the peer-support chatbot's local sentiment engine identifies explicit high-risk phrasing (such as keywords related to self-harm or ending one's life), it immediately interrupts normal operations. It applies a distinct red visual alert to the message bubble, switches the chat dialogue to emergency support mode, and displays a prominent, unmissable call-to-action urging the student to launch the global SOS Emergency dialog.

4.10. Inferred Database Architecture and Platform Settings

The background architecture utilizes a cloud-hosted, relational database configuration composed of five core tables to manage the application's workflows smoothly:

- **profiles Table:** Links to the primary user authentication records, housing account registration flags (`is_registered`), profile metadata, and automated anonymous identifier strings.
- **therapists Table:** Acts as the master registry for participating campus counselors, detailing names, specialization tags, professional bios, photo paths, ratings, and active statuses.
- **therapist_availability Table:** Manages weekly recurring calendar schedules for counselors, charting day designations, 24-hour time ranges, and permissible session formats (online vs. offline).
- **appointments Table:** Logs confirmed bookings, linking the tracking data, chosen counselors, scheduling blocks, custom user notes, and auto-generated meeting rooms or physical office addresses.

- **support_cards Table:** Securely stores pre-session emotional ratings, mood sliders, and written narratives, connecting them directly to upcoming appointments.

The user interface is built for modern accessibility standards, utilizing an elegant layout theme with customizable design values configured inside a global stylesheet. To protect student tracking preferences in full compliance with GDPR mandates, the integrated `CookieConsentBanner` saves granular cookie choices locally to the device's storage. Furthermore, development safety configurations completely isolate production web tracking modules (such as Vercel Analytics) from running during local testing or building phases, ensuring no student activity data is accidentally leaked during platform updates.

5. Summary and conclusions

5.1. Project Summary

- The PsychSupport platform was created as an institutional response to the growing mental health crisis among university students, providing a secure, low-barrier digital ecosystem for mental health care. By utilizing a fluid single-page setup, the application completely eliminates loading delays, ensuring an uninterrupted user experience for students in moments of acute distress.
- The core philosophy of PsychSupport is to bridge the gap between immediate student privacy and the administrative needs of university counseling services. Through a dual-tiered access model, students can instantly enter a hidden, anonymous layer to use calming tools and interactive chat features without revealing their identity. When ready, they can seamlessly upgrade to a registered profile to unlock direct counselor bookings and pre-appointment diagnostics. Over its development cycle, the platform has successfully evolved from a conceptual framework into a secure, production-ready environment capable of supporting a university community.

5.2. Met Problems and Implemented Solutions

During the development and system integration, the team encountered several major operational and design challenges. Each hurdle was resolved through strategic functional solutions:

- **The Anonymity vs. Accountability Dilemma:** Providing immediate access while maintaining safety tracking without real personal data was a primary challenge. *Solution:* The team implemented an automatic background account generation script. When entering the anonymous tier, the system sets up a randomized secure profile, keeping login data entirely hidden from the student and exposing only a clean, private identifier (e.g., ANON-XK7F2A).
- **Student Privacy Boundaries in Communication:** Sending raw student chat logs to external cloud text analysis tools would violate data privacy laws and erode student trust. *Solution:* The team developed an isolated sentiment detection engine that executes entirely within the user's local browser window. It analyzes text locally to gauge the user's emotional state, ensuring absolute conversation confidentiality.
- **Immediate Crisis Safety Risks:** A standard navigation menu could delay access to critical phone numbers if a student experienced a sudden panic attack. *Solution:* A global emergency safety net (SosDialog) was embedded permanently into the sidebar footer layout, keeping it visible on every screen view. Furthermore, if the chat engine detects explicit high-risk keywords regarding self-harm, it automatically overrides standard messaging and displays an unmissable call-to-action to open the emergency panel.
- **Scheduling Coordination and Appointment Context:** Coordinating flexible counseling times across different formats (virtual vs. on-campus) without data fragmentation proved difficult. *Solution:* The team built an adaptive booking system across a six-day grid that generates unique video rooms for online care or provides exact room locations for in-person visits. For on-campus visits, it links a digital "Support Card" (mood sliders and text

descriptions) directly to the appointment, letting counselors review the student's emotional context before the session begins.

- **Session Vulnerability and Timeout Disruptions:** Relying on basic browser validation could lead to session hijacking or sudden logouts, disrupting a student during a sensitive counseling interaction. *Solution:* A server-driven middleware script was deployed to intercept network transitions, continuously refreshing user sessions behind the scenes and locking down student data the moment an invalid session is detected.

5.3. Unsolved Issues and Constraints

While the primary operational framework of PsychSupport is fully realized, a few practical constraints remain due to the initial scope boundaries of the project:

- **No Multi-Device Chat Synchronization for Anonymous Users:** Because anonymous account details are strictly isolated to the specific browser storage where they were created, a student cannot view or continue an anonymous chat history if they log in from a different device or clear their browser cache.
- **Static Content Updates for the Knowledge Library:** To ensure instant article loading speeds, the psychoeducational materials are embedded directly within the platform's core layout. While this removes loading delays, it means updating or adding new articles requires a code deployment rather than simple database entries updates.
- **External Video Consultation Handling:** The platform generates unique, secure video links for online therapy sessions, but the actual video and audio interactions take place via external browser redirections rather than being natively embedded directly within the dashboard interface.

5.4. Possibilities for Further Development

To build upon the successful core of the platform, several future enhancements have been mapped out:

- **Dynamic Content Management System (CMS):** Transitioning the static article library into a live CMS will allow university mental health professionals to write, edit, and publish new articles instantly without needing technical support.
- **Native Video Consultations:** Integrating video and audio communication directly into the dashboard using secure web protocols will allow students to conduct online therapy sessions entirely within the secure shell of PsychSupport.
- **Encrypted Notification System:** Adding a secure push notification worker will let the platform send real-time appointment reminders and chat notifications without compromising anonymous browser states.
- **Anonymous Account Recovery Keys:** Future updates will introduce an option for anonymous users to download a single-use encrypted key file, enabling them to safely recover their chat history and logs on other devices without ever inputting real personal information.
- **Cross-Institutional Scaling:** The architecture can be scaled into a multi-tenant framework, allowing other universities to deploy independent instances of PsychSupport tailored to their specific campus layouts, local counseling directories, and emergency numbers.

5.5. Team Members Contribution

The implementation of the PsychSupport platform is the result of a coordinated group effort, with each member taking ownership of specific functional layers of the application ecosystem:

- **Joanna Joniec (Project Lead)** Joanna handled the overall project vision, guiding the structural development from inception to completion. She was responsible for shaping PsychSupport into a student-centered solution, managing team alignment, and ensuring that all built functionalities matched the real-world mental health and support needs of the student user base.
- **Marcin Gronek (Technical Development)** Marcin focused on full-stack implementation, backend architecture, and Supabase database integration. He took full ownership of the project's technical documentation, authoring the structural blueprints, database definitions, and user guides. He designed the relational database schema, mapping out the tables for user profiles, counselor listings, appointment slots, and emotional support cards, while implementing server-side session flows and backend token handling.
- **Szymon Gustek (Product Design & Communication)** Szymon managed the user interface layout and product user experience across the application. He ensured the single-page platform complied with standard accessibility and web contrast requirements. Szymon translated mental health workflows—such as the evidence-based calming guides, peer chat window, and therapist booking calendar—into an accessible, clean, and intuitive user experience designed to reduce cognitive load.

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